

Exploring Simulated Web Agents vs Real User Journeys

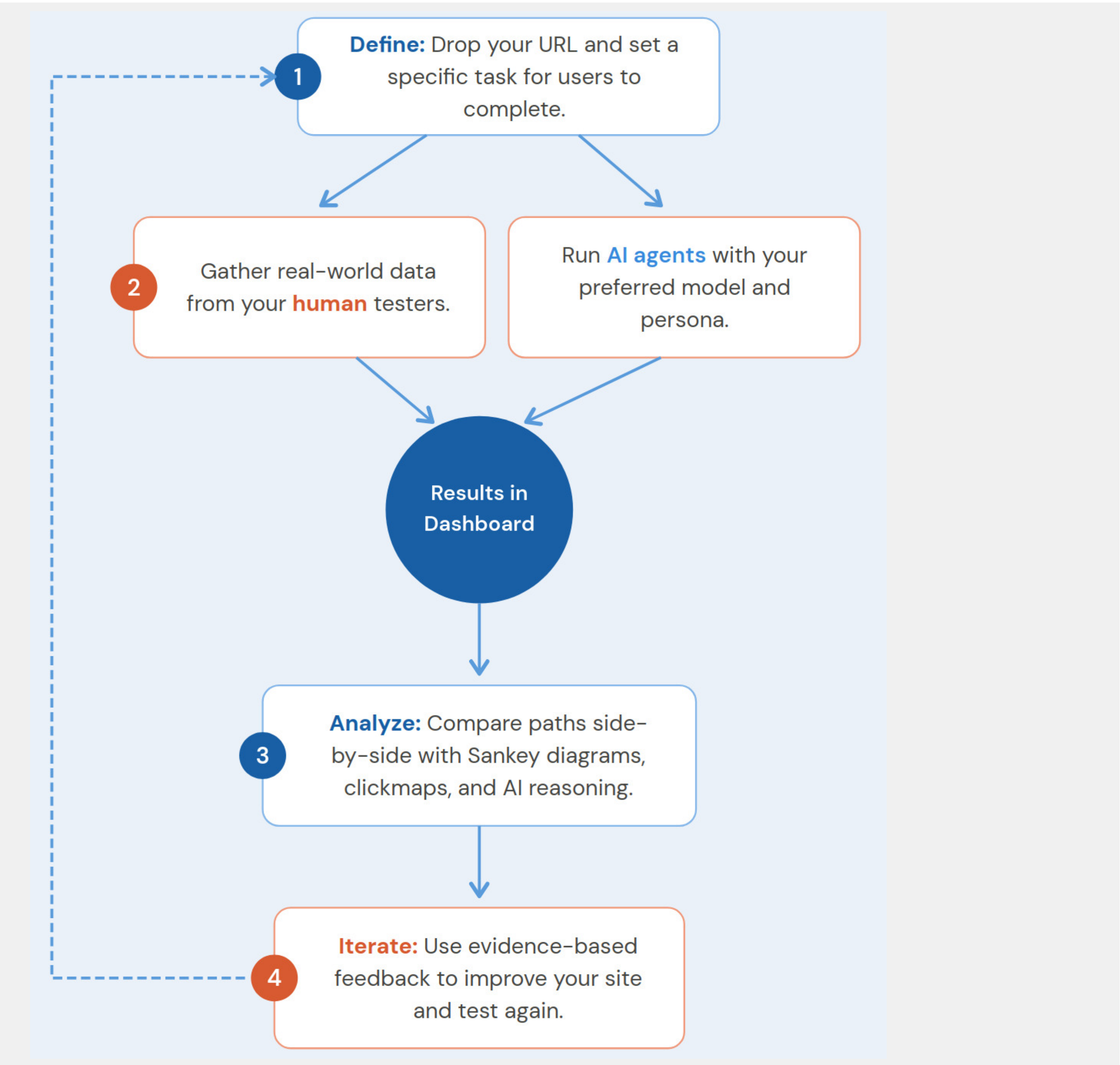
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1 Introduction

We propose a framework for UX designers to scale UX evaluation, replacing the slow data collection loop with aligned web agents.

- **Scale UX evaluation**
Reduce slow human testing cycles by running AI agents on the same tasks in parallel.
- **Compare AI vs. human navigation**
Identify where agents and users diverge, surfacing friction points that take weeks to uncover manually.
- **Human-in-the-loop re-evaluation**
Make changes, re-run agents. Version-stamped journeys enable before/after comparison.

2 Workflow



3 Users & Data

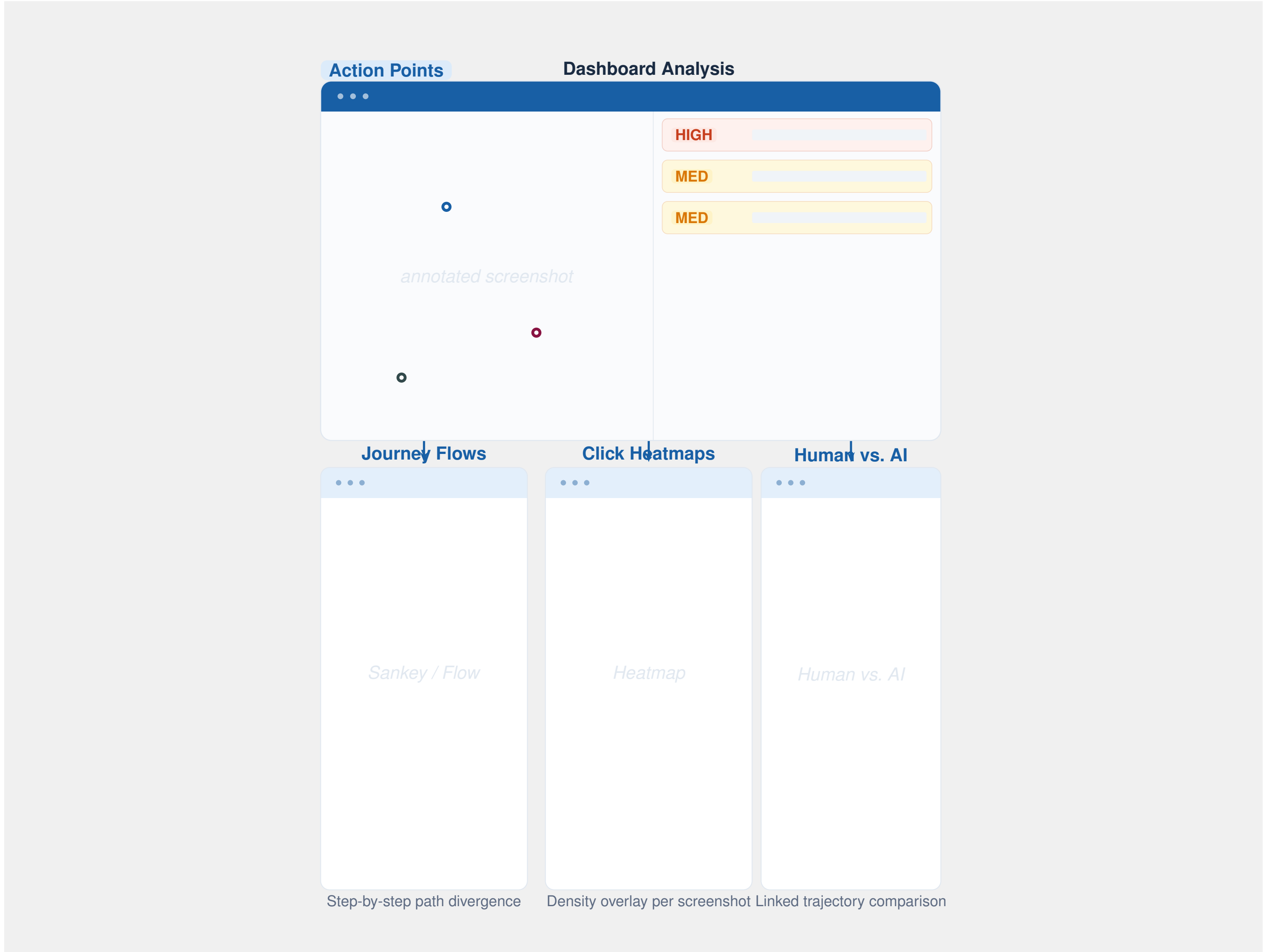
Users

- **UX designers & Web developers**
Evaluating and improving web interfaces. Validating navigation flows and detecting usability issues.

Data

- **Human User Journeys**
Collected via reverse proxy distributed to real testers. Every click, scroll, and navigation recorded per task.
- **Web Agent Journeys**
AI browser sessions running the same tasks. Full step history with per-step screenshots.
- **Collected per step**
Clicks · Scrolls · Screenshots · Timestamps · URLs · Task descriptions

4 User Interface



5 Agent Steering & Evaluation

Analysis & Dashboard

- **Action Points**
Task screenshots are annotated with glyphs marking where issues occur. Each glyph links to an LLM-generated pain point with human and agent explanations and a matched recommendation. Each action point routes to the most relevant visualisation for deeper analysis.

- **Visualizations**

Journey Flows (Sankey) · Click Heatmaps · Human vs. AI trajectory view. Filterable by task and journey source.

Agent Steering & HITL Re-evaluation

- **Alignment**
Recorded human journeys are converted into prompt injections that steer agents toward human-like navigation patterns, helping agents overcome complex scenarios. Steered and unsteered trajectories are compared in the Flow Chart view.

- **Re-evaluation**

Recorded human journeys enable re-evaluation via direct prompt-injection, mitigating the need for manual evaluation again and scaling up the process.

6 Conclusion

AI-driven automation benefits UX evaluation but requires alignment with human expectations. Our HITL framework reduces the manual testing bottleneck while maintaining behavioral fidelity through trajectory-guided steering. Agents are evaluated both with and without human steering, enabling measurement of alignment quality. A remaining limitation is agent robustness on complex, dynamic webpages. Future work targets tighter feedback loops and automated change propagation from action points to implementation.